

ASIM ASHFAQ

Digital Asset Infrastructure | Compliance Engineering | Audit

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PROFILE

I build regulated digital-asset infrastructure, and I now examine it from the other side. As Co-Founder and CTO of [0xEquity](#) I decide what we build and under whose rules: tokenised real estate and equities, fractional positions from small tickets, rental income paid on-chain, live on Base with Arbitrum in rollout. As Head of Digital Assets and Blockchain Technologies at AKSİS, an independent audit firm authorised by Turkey's KGK, Capital Markets Board and banking regulator, I head the practice working inside the regime that requires every Turkish crypto-asset service provider to commission an annual independent information systems audit.

The harder half of the building is compliance, not code. Everything we issue goes out under Regulation S, so I built the eligibility path behind it: a server-side non-US-person determination under Rule 902(k), versioned, evidenced, and written to a tamper-evident ledger before the decision is stored, with treaty withholding resolved server-side rather than accepted from the client. Our consumer application went through repeated App Store and Play review cycles under the named cryptocurrency, exchange and financial-trading guidelines, and is live on both stores. The question was rarely whether the code ran. It was whether an activity fell inside a perimeter, and whether we could show where the line sat.

The systems underneath are mine to the code: a NestJS backend of some fifty domain modules, a Go execution stack on durable workflows with an MCP server that lets a language model author and run on-chain strategies under keys that cannot move funds, a Groth16 proving stack in Rust, and the zero-trust service mesh all three sit on. Twelve years across four operating cultures: a US-incorporated company run from Istanbul, a Turkish audit firm, a German proptech, a Thai retail platform, and Pakistani teams in Islamabad.

EXPERIENCE

Head of Department, Digital Assets and Blockchain Technologies

AKSİS Uluslararası Bağımsız Denetim A.Ş. | Istanbul | 2026 – Present

Head of the digital assets practice at an independent audit firm authorised by Turkey's KGK, the Capital Markets Board (SPK) and the banking regulator (BDDK). Turkey's 2024 crypto-asset law and the CMB's 2025 communiqués require every crypto-asset service provider to commission an annual independent information systems audit from a CMB-authorised firm, which is the regime this practice works inside.

- Led the team that built the firm's proof-of-reserve system: address discovery, multi-chain balance aggregation and reserve verification across more than 200,000 addresses on over 100 blockchains.
- Ran the Bybit engagement, a compliance audit covering 2025 activity.
- Brought the firm's group audit methodology (ISA 600 scoping, component auditor instructions, independence confirmations and engagement partner sign-off) to bear on digital asset engagements.
- Translate on-chain evidence into assurance work that a regulator and an audit committee can both read.

[Proof of reserve](#) · [CASP audit](#) · [SPK / CMB](#) · [ISA 600](#) · [Multi-chain](#) · [AML](#)

Co-Founder & CTO

0xEquity US, Inc. | Remote | Nov 2022 – Present

- Founded and built the engineering side of a real-world-asset tokenisation platform for tokenised real estate and equities, live on Base with Arbitrum in rollout.
- Wrote and maintain the NestJS/TypeScript backend: roughly fifty domain modules covering KYC/AML, orders, trades, a match-settlement engine, holdings, rewards, fiat on/off-ramp, campaigns and relayed on-chain transfers.
- Designed the platform's zero-trust infrastructure: SPIFFE/SPIRE workload identity issuing mTLS to every service, HashiCorp Vault for secrets and signing keys, Falco for runtime detection, and immudb as a tamper-evident audit ledger.
- Moved long-running, externally-settled operations onto Temporal, so cross-chain bridge steps, unfilled limit orders and transactions needing fee bumps survive restarts instead of being lost mid-flight.
- Built authentication around passkeys and OPAQUE rather than transmitted passwords, with Casbin policy for authorisation and TOTP as a second factor.
- Built the React dApp on wagmi/viem (trading, portfolio, lending and a pro order ticket) with a swap router that reports real price impact rather than an unmeasured zero.
- Built the investor-eligibility path for a Regulation S offering: a server-side non-US-person determination under Rule 902(k) that records which input decided it and at what assurance level, because a self-declared nationality is weaker evidence than a verified identity and the audit basis has to say so. Every determination is anchored to a tamper-evident ledger before the row is persisted, so no decision can exist without its trail.
- Resolved US withholding server-side instead of trusting the client with it: a W-8BEN treaty resolver covering thirty jurisdictions that derives the article and the rate from the investor's treaty country, the income type and the issuing entity's tax election, checked against IRS Table 1, Publication 515 and IRC section 897, with REIT ownership thresholds handled per treaty.
- Shipped the React Native mobile app on both iOS and Android, from passkey sign-up and KYC through fiat on-ramp, trading and rent claims, and took it through App Store and Play review.
- Started DeFiLoops, the platform's agentic execution layer: a Go stack on durable workflows that turns a strategy described in one sentence into a numbered plan, approved once and then run unattended across networks.

- Designed its key model to hold on-chain rather than by policy: the agent signs typed adapter parameters with no field for a destination or raw calldata, is capped by a spend limit it cannot raise, and is revocable across a whole chain in one transaction. An MCP server exposes plan authoring to a language model, with the money-spending tools withheld at the server rather than by convention.
- Run the whole delivery loop solo: reproducible Nix dev shells, a Dockerised local stack, CI, and Prometheus/Grafana/Metabase for what happens after deploy.

[NestJS](#) · [TypeScript](#) · [Go](#) · [React](#) · [Temporal](#) · [PostgreSQL](#) · [SPIFFE/SPIRE](#) · [Vault](#) · [Solidity](#) · [MCP](#) · [React Native](#) · [Regulation S](#) · [W-8BEN](#)

Full Stack Developer

Jarvis OOD | Remote | Mar 2020 – 2025

- Front-end work on Jarvis Network, a set of protocols connecting traditional finance to DeFi. Synthetium issues synthetic fiat assets (jEUR, jCHF and other jFIATs) priced by Chainlink oracles, governed by the Jarvis DAO and its JRT token.
- Built the platform's interfaces in React and Next.js.

[React](#) · [Next.js](#)

DevOps / Full Stack Developer

EverReal GmbH | Remote | Oct 2020 – Aug 2021

- Worked on EverReal, a Munich proptech running a cloud-based white-label platform that digitises the whole rental and sales transaction: listing and marketing, applicant management, viewing appointments, electronic contracts and handover.
- Split between application development and the infrastructure the B2B SaaS platform ran on.

[SaaS](#) · [DevOps](#)

DevOps / Full Stack Developer

WeFresh, Ascend Commerce | Remote | Dec 2019 – Aug 2020

- Worked on WeFresh, Ascend Commerce's Bangkok grocery and convenience delivery app, which fulfils orders through 7-Eleven, CP Freshmart and Makro wholesale.
- Split between backend service work and the delivery and deployment infrastructure behind it.

[Node.js](#) · [DevOps](#)

Team Lead

Expanse.tech | Islamabad | Aug 2016 – 2019

Led a team of ten engineers building the Expanse blockchain ecosystem, with several products running at once rather than one after another, on stacks that shared almost nothing and for audiences that had little in common: a public explorer, a token issuance platform for founders raising money, and the desktop wallet ordinary holders used.

- Gander, the chain's block explorer and dApp analytics tooling: Golang REST APIs behind a React front end, realtime workers pulling node data into a persistent store, and Elasticsearch indexing it for search.
- Tokenlab, the token issuance and ICO platform: a modular Solidity contract system whose sale contracts carried time-locked distribution so investors were not exposed to a founder dumping supply, deployable to either Expanse or Ethereum.
- Luna Wallet, the cross-platform desktop wallet most holders met the chain through, with Tokenlab folded into version two so a holder could reach a sale without leaving the wallet.
- Split ten people across three concurrent products and five stacks (Golang services, Vue and React front ends, Electron desktop, Solidity contracts), and kept all three shipping.

[Team of 10](#) · [Golang](#) · [Vue](#) · [React](#) · [Electron](#) · [Solidity](#) · [Elasticsearch](#)

Freelance Developer

Freelancer.com | Islamabad | 2012 – Mar 2017

- Web and software development for clients on Freelancer.com, working there since 2012: WordPress and PHP/MySQL builds, e-commerce and payment gateway integration, and third-party API work across PayPal, Facebook and Google.
- Rated 4.7 out of 5 across 72 reviews, with 88% on-time delivery, 94% on-budget completion, and 39% of clients hiring again.
- Larger engagements included a SaaS clinic management system (Angular 2, MEAN) and a realtime chat/video application (Node.js, MQTT, STOMP, RabbitMQ).

[Angular 2](#) · [MEAN](#) · [Node.js](#) · [MQTT](#) · [RabbitMQ](#)

PROJECTS

0xEquity (0xequity.com)

2022 – present

A platform for investing in tokenised real-world assets (property and equities) and putting them to work on-chain. Fractional positions from small tickets and rental income paid on-chain.

- NestJS backend of ~50 domain modules, from KYC and document signing through orders, settlement, holdings and rewards.
- Live on Base with Arbitrum in rollout, behind an RPC proxy that spends a read's retries across different providers rather than failing on one.
- Zero-trust service mesh: SPIRE-issued SPIFFE identities, mTLS between every service, Vault-held signing keys, immudb audit ledger, Falco runtime detection.

- Durable Temporal workflows for anything that settles outside a request: bridge legs, resting orders, transaction bumps.
- React dApp on wagmi/viem with RainbowKit, TanStack Query, Zustand and i18n.

[NestJS](#) · [TypeScript](#) · [React](#) · [wagmi/viem](#) · [Temporal](#) · [PostgreSQL](#) · [TypeORM](#) · [Redis](#) · [RabbitMQ](#) · [MQTT](#) · [gRPC](#) · [SPIFFE/SPIRE](#) · [Vault](#) · [Falco](#) · [immudb](#) · [Casbin](#) · [WebAuthn](#) · [Docker](#) · [Nix](#)

0xEquity mobile app (iOS and Android)

[React Native](#) | shipped to the App Store and Play

The mobile client for the platform: passkey sign-up, KYC, a smart-contract wallet deployed at first transaction, fiat on-ramp, property trading and rent claims, gasless throughout because signed transactions go to a relay that pays the gas.

- React Native with Expo, one codebase across both platforms, with native release pipelines through Fastlane and EAS.
- WebAuthn passkeys and biometric keychain for sign-in, social recovery through three trusted guardians, and device attestation via Play Integrity and App Attest.
- Took the app through Apple review against the guidelines that govern financial and crypto products: 3.1.5(a)(iii) on cryptocurrency, 3.1.5(a)(iv) on exchanges and 3.2.1(viii) on facilitating trades.

[React Native](#) · [Expo](#) · [TypeScript](#) · [viem](#) · [WebAuthn](#) · [TanStack Query](#) · [Fastlane](#) · [EAS](#) · [Maestro](#) · [Sentry](#)

DeFiLoops ([defiloops.com](#))

[Go](#) | agentic execution

The execution layer for multi-step DeFi. You describe a strategy in a sentence; it becomes a numbered plan you approve, and then runs for hours or days across networks, on an account only you can empty.

- Six-stage compiler from conversation to signed transaction, where each stage narrows what is possible and any stage can refuse.
- Steps that belong together are merged into one transaction, so an approval cannot outlive the step it was granted for; a step can spend what the previous step produced.
- Atomic loan migration via flash loan: a deadlock by hand, one instruction here, with the borrowed funds never touching the user's account.
- Key model enforced on-chain rather than by policy: the agent key signs typed adapter parameters with no field for a destination or raw calldata, is capped by a spend limit it cannot raise, and is revocable for every account on a chain in one transaction.
- Own swap router answering questions an aggregator cannot: what fills at a named limit price, and how much a resting order takes right now.
- Durable runs on Temporal, plus schedules, price triggers and plan chaining that rebuild and re-price the plan each time they fire.
- Foundry invariant tests for the adversarial cases: agent blast radius, guard drain, proxy squatting, address parity.

[Go](#) · [Temporal](#) · [Solidity](#) · [Foundry](#) · [ERC-4337](#) · [PostgreSQL](#) · [go-ethereum](#) · [Uniswap v3](#) · [Aave](#) · [CCTP](#)

MCP server for agent-driven DeFi

part of DeFiLoops

A Model Context Protocol server that lets a language model author, validate and run multi-step on-chain strategies, built to find out whether a small open model could write a plan the system would accept, and to make the answer yes.

- 22 tools in five groups (catalogue, chain reads, plan authoring, chains and schedules, and run watching) over ~33k lines of Go with 84 test files.
- Two transports with different guarantees: stdio, where a pipe's single reader makes unexposed tools structurally unreachable; and a loopback socket requiring a bearer token, refusing to bind anything but localhost.
- Tools that spend money are withheld at the server unless explicitly enabled. On a pipe the client's request is a guarantee; on a socket it is only a preference.
- Validation goes over the wire to the running API rather than importing the compiler, so the model validates against the build that will actually execute it.
- Per-request credential binding on a value receiver, so two concurrent conversations cannot swap identities under load.
- 21 written skill guides the agent reads before answering, held to one rule: a guide describes what the code does today, or it starts lying.

[MCP](#) · [Go](#) · [JSON-RPC 2.0](#) · [LLM tool use](#) · [OpenRouter](#)

rapidzkey ([github.com/asimashfaq/rapidzkey](#))

[Open source](#) | MIT

Fast Groth16 trusted setup for Circom circuits, a drop-in replacement for `snarkjs groth16 setup` producing byte-identical output.

- A ~2M-constraint circuit completes in about three minutes on eight cores, against a snarkjs baseline still unfinished when it was stopped at fourteen minutes.
- Accumulates weighted curve points by scatter-add across Rayon threads, rather than running a separate MSM per signal.
- Computes missing Lagrange-basis points by parallel IFFT, so unprepared ptau files work without a separate preparation pass.

[Rust](#) · [Rayon](#) · [arkworks](#) · [Groth16](#) · [BN254](#) · [Circom](#)

zk-realestate

[Rust](#) | zero-knowledge

[Rust](#) · [arkworks](#) · [Circom](#) · [WASM](#) · [napi-rs](#) · [axum](#) · [BLS](#) · [WebAuthn](#)

Content Engine

Python | Dagster | OxEquity

Python · Dagster · Claude API · MCP · FastAPI · PostgreSQL · Whisper · Vault · Nix

Indexer-core

TypeScript / Bun | Bazel

TypeScript · Bun · Bazel · Nix · Dagster · PostgreSQL · Foundry

Jarvis Network (jarvis.money)

2020 – 2025

React · Next.js · Solidity · Chainlink · Ethereum · Polygon · Synthetic assets · AMM routing

Full Stack Starter (github.com/asimashfaq/fullstack-starter)

Open source

NestJS · Next.js · gRPC · GraphQL · nx · TypeScript · RavenDB · Storybook · Envoy · Docker

Gander (gander.tech)

Expanse.tech | 2016 – 2019

Golang · React · Redux · Elasticsearch · Couchbase · MongoDB · Redis · WebSockets · JSON-RPC · AWS

Tokenlab (tokenlab.io)

Expanse.tech | 2016 – 2019

Vue · Golang · Node.js · Solidity · Truffle · Ganache · Mocha · web3.js · HD Wallet · Meta transactions

Luna Wallet (github.com/expanse-org/luna-wallet)

Expanse.tech | Open source

Vue · Electron · Blockchain node · ABI · REST API

Other earlier work

Clinic Management System: SaaS on Angular 2 and the MEAN stack. **Random Chat / Video App:** Node.js, MQTT, STOMP, RabbitMQ. **Places App** and **TicTac App:** React, on GitHub.

SKILLS

AI tooling	Model Context Protocol · LLM tool-use design · Agent skill authoring · Claude API
Languages	TypeScript / JavaScript · Go · Rust · Python · Solidity · SQL
Backend	NestJS / Node.js · Temporal (durable workflows) · Dagster (data orchestration) · REST / GraphQL · gRPC · RabbitMQ / Kafka / MQTT · WebSocket / realtime PubSub · Microservices
Frontend	React / Next.js · React Native / Expo (iOS, Android) · wagmi / viem / RainbowKit · TanStack Query / Zustand · Tailwind CSS · Vue / Angular / Electron
Data	PostgreSQL / TypeORM · Redis · NoSQL / Elasticsearch · immudb / Hasura
Blockchain & Web3	ERC-4337 account abstraction · Foundry / Hardhat · EIP-2535 Diamonds · DeFi: AMMs, lending, flash loans · Cross-chain bridging (CCTP) · RWA tokenisation · Proof of reserve · Regulation S / Rule 902(k) · W-8BEN treaty withholding
Cryptography	Groth16 zkSNARKs / arkworks · Circom circuits · WebAuthn / passkeys · OPAQUE / BLS signatures
Infrastructure	Kubernetes / k3s / Rancher · Docker / Nix / Helm / Istio · Bazel / Bun · Terraform / GitOps · SPIFFE / SPIRE / Vault · Falco / Open Policy Agent · Prometheus / Grafana · GitLab CI / CircleCI

EDUCATION

BS Software Engineering
Riphah International University, Islamabad
2010 – 2014

I.Com
Punjab College of Commerce, Rawalpindi
2005 – 2007

HSSC
Fazaia International School, Rawalpindi
2003 – 2004

VOLUNTEERING

Founder, “Pukaray Pakistan”
Social organisation, Islamabad
2009 – 2015

Volunteer, IEEE Pakistan Mission to Progress
IEEE Pakistan Women in Engineering Forum,
Lahore
2012 – 2013

Volunteer, Aghosh Foundation
Attock
2009 – 2010

LANGUAGES

English (Professional) · Urdu (Native)

QUOTE

“Shoot for the moon. Even if you fail, you will fly among the stars.”



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